# CMSC 420: Extended KD Trees

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**1 Problems** **2**

# 1 Problems

1. In an extended KD-tree when we split according to the largest spread we use the median of the node's data. What is a good reason not to use the mean?

2. Consider the following extended kd-tree. The tree is on the left and the plane diagram is on the right. The shaded boxes are the bounding boxes for the leaf nodes.

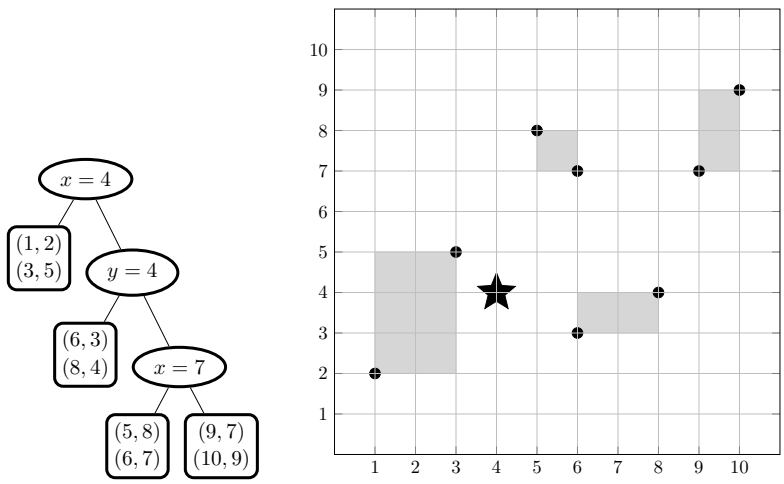


Figure 1: An EKD Tree and Some Bounding Boxes

(a) Draw separate plots of the bounding boxes for each non-leaf node.

(b) Suppose we are looking for the two nearest neighbors to $(4, 4)$. We are following the algorithm discussed in class and used on the coding project. Walk through this algorithm, explaining each step briefly, and write down what the list looks like after each change

3. Give a graphical example in 2D of a point $P$ and two bounding boxes $B_1$ and $B_2$ such that all of the following are satisfied:

   - $P$ is not in either bounding box.
   - Each bounding box contains exactly two points.
   - $P$ is closer to $B_1$ than it is to $B_2$.
   - $P$ is closer to both points in $B_2$ than it is to both points in $B_1$.

4. Consider the following extended kd-tree. The tree is on the left and the plane diagram is on the right. The shaded boxes on the right are the bounding boxes for each leaf. The bounding boxes for the (non-leaf) subtrees are not shown.

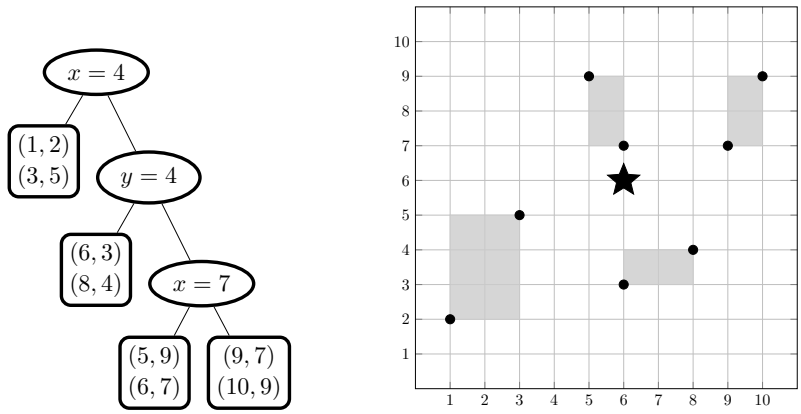


Figure 2: An EKD Tree and Some Bounding Boxes

Suppose we are looking for the three nearest neighbors to $(6, 6)$.

We are following the algorithm discussed in class and used on the coding project. For each of the following points determine whether the point will ever be on our list (even if it's removed later) and whether it will be on our list when the algorithm completes. Assume that when leaf lists are checked they are checked in the order the points are given in the tree (top-down). Fill in each space with either **Y** or **N**.

| **Point** | **Ever in List?** | **In List at End?** |
|---|---|---|
| $(1, 2)$ | | |
| $(3, 5)$ | | |
| $(6, 3)$ | | |
| $(8, 4)$ | | |
| $(6, 7)$ | | |
| $(5, 9)$ | | |
| $(9, 7)$ | | |
| $(10, 9)$ | | |

5. Assuming an EKD-tree with $k = 2$ and $m = 1$ using cycle splitting, give an example of a sequence of four insertions followed by one deletions which results in the tree having exactly two splitting nodes both of which split by $x$. Draw sufficient pictures to justify your process and result.

6. Consider the following extended kd-tree. The tree is on the left and the plane diagram is on the right. The shaded boxes on the right are the bounding boxes for each leaf. The bounding boxes for the (non-leaf) subtrees are not shown.

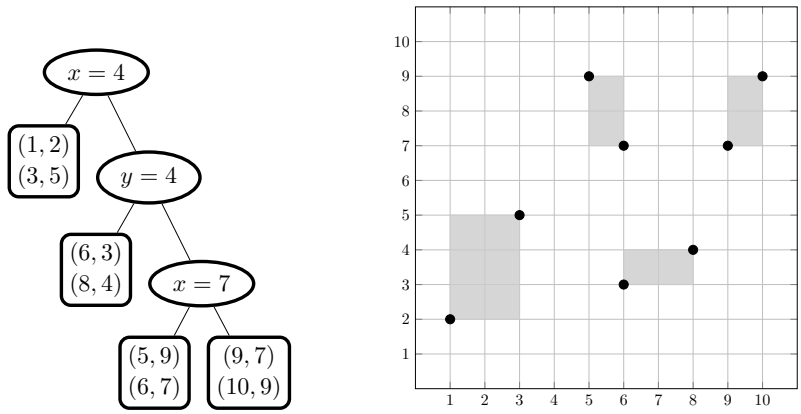


Figure 3: An EKD Tree and Some Bounding Boxes

Suppose we are performing a range query for points in the query box:

$$Q = [2, 9] \times [3, 8]$$

We are following the algorithm discussed in class except for a slight modification (also discussed in class) where - if we need to visit both subtrees of a splitting node we should first visit the subtree whose bounding box has larger overlap with $Q$, then the other one.

(a) In what order would the leaves be visited? Only consider those which would actually be visited.

(b) What points would be in the list at the end?

7. Consider the following extended kd-tree. The tree is on the left and the plane diagram is on the right. The shaded boxes on the right are the bounding boxes for each leaf. The bounding boxes for the (non-leaf) subtrees are not shown.

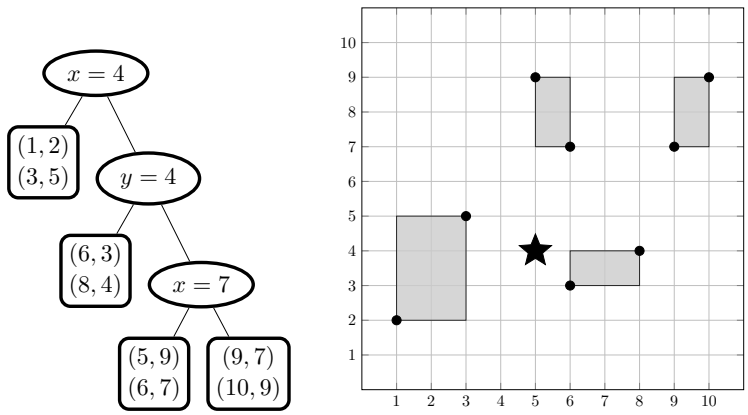


Figure 4: An EKD Tree and Some Bounding Boxes

Suppose we are looking for the three nearest neighbors to $(5, 4)$.

We are following the algorithm discussed in class and used on the coding project. For each of the following points determine whether the point will ever be on our list (even if it's removed later) and whether it will be on our list when the algorithm completes. Assume that when leaf lists are checked they are checked in the order the points are given in the tree (top-down). Fill in each space with either **Y** or **N**.

Note: Please do not use any other symbols, and empty boxes will be marked as incorrect.

| **Point** | **Ever in List?** | **In List at End?** |
|---|---|---|
| $(1, 2)$ | | |
| $(3, 5)$ | | |
| $(6, 3)$ | | |
| $(8, 4)$ | | |
| $(6, 7)$ | | |
| $(5, 9)$ | | |
| $(9, 7)$ | | |
| $(10, 9)$ | | |

8. Consider the following extended kd-tree. The tree is on the left and the plane diagram is on the right. The shaded boxes on the right are the bounding boxes for each leaf. The bounding boxes for the (non-leaf) subtrees are not shown. Notice also that the leaf nodes have labels $A, B, C, D$.

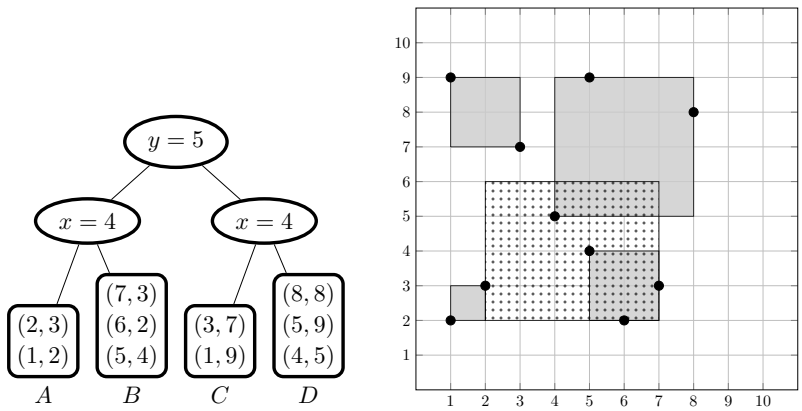


Figure 5: An EKD Tree and Some Bounding Boxes

Suppose we are performing a range query for points in the query box:

$$Q = [x_{min}, x_{max}] \times [y_{min}, y_{max}] = [2, 7] \times [2, 6]$$

We are following the algorithm discussed in class except for a slight modification (also discussed in class) where - if we need to visit both subtrees of a splitting node we should first visit the subtree whose bounding box has larger overlap with $Q$, then the other one.

(a) In which order are the leaf nodes visited?

| Order | |
|---|---|

(b) In the following table identify the points which will be in our list. Fill in each space with either **Y** or **N**.

| **Point** | **In List?** |
|---|---|
| $(2,3)$ | |
| $(1,2)$ | |
| $(7,3)$ | |
| $(6,2)$ | |
| $(5,4)$ | |

| **Point** | **In List??** |
|---|---|
| $(3,7)$ | |
| $(1,9)$ | |
| $(8,8)$ | |
| $(5,9)$ | |
| $(4,5)$ | |

9. In an EKD tree with $m = 4$ we have an overfull node containing these points:

$$(2, 1, 1), (5, 2, 6), (0, 4, 10), (1, 3, 0), (1, 1, 1)$$

If we perform a spread split on the points, fill in the splitting node data and leaf points below. The order of the points in the leaves doesn't matter.

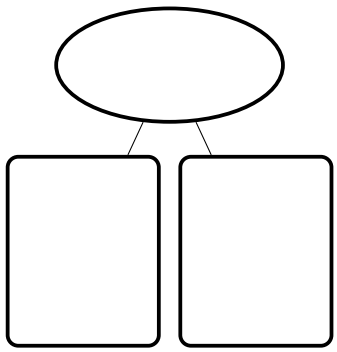

Figure 6: Spread Split Result

10. Given the following set of points:

$$(1,2,3,4),(3,1,2,4),(2,1,3,4),(4,1,3,2),(5,1,3,2),(5,2,3,5)$$

(a) Cycle sort them by the first coordinate.

(b) Cycle sort them by the second coordinate.

(c) Cycle sort them by the third coordinate.

(d) Cycle sort them by the fourth coordinate.

11. Suppose we start with an empty EKD tree with $m = 2$ and use cycle splitting.

    (a) Show the final result after inserting the points $(1, 3, 2)$, $(1, 2, 3)$, and $(5, 2, 3)$.

    (b) Continuing from the above, show the final result after inserting the points $(0, 1, 1)$ and $(2, 5, 0)$.

12. Suppose we start with an empty EKD tree with $m = 2$ and use spread splitting.

    (a) Show the final result after inserting the points $(0, 2, 7)$, $(3, 1, 1)$, and $(1, 1, 2)$.

    (b) Continuing from the above, show the final result after inserting the points $(9, 1, 0)$ and $(0, 0, 1)$.